SOFTWARE REQUIREMENTS SPECIFICATION

Customer Churn Analysis

Excel Workbook — IBM Telco Dataset

Document Type	Software Requirements Specification (SRS)
Project	Customer Churn Analysis — Excel
Dataset	IBM Telco Customer Churn Dataset (Kaggle)
Tool	Microsoft Excel
Author	Laxman Sharma
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Status	Final — Portfolio Submission

1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for the Customer Churn Analysis project using Microsoft Excel. It documents the objectives, scope, stakeholders, functional requirements and constraints for analyzing customer churn from the IBM Telco dataset. The document is intended for academic evaluation and portfolio review purposes.

1.2 Project Background

Customer churn in the telecom industry represents a significant revenue risk. This project uses Microsoft Excel to analyze churn patterns across the IBM Telco Customer Churn Dataset (7,043 records), uncovering key drivers such as contract type, tenure, payment method and service usage. Findings are presented through interactive dashboards and pivot-based analysis.

1.3 Scope

In scope:

- Data import, cleaning and transformation in Microsoft Excel
- Exploratory data analysis using Excel pivot tables and formulas
- Visualization using Excel charts and conditional formatting
- KPI calculations and interactive dashboard
- Documentation of data dictionary and transformation log

1.4 Definitions

Churn	A customer who has cancelled or discontinued their telecom service
Churn Rate	Percentage of customers who churned in the dataset
KPI	Key Performance Indicator — a measurable value tracking business outcome
Pivot Table	Excel feature that summarises and aggregates data interactively
IBM Telco Dataset	Public dataset of 7,043 telecom customers with 21 attributes, sourced from Kaggle

2. OVERALL DESCRIPTION

2.1 Product Perspective

This is a standalone Excel workbook that serves as an end-to-end analytical report. The workbook ingests raw CSV data, applies transformations, performs pivot-based analysis and presents findings through a management-ready dashboard, all within a single .xlsx file

2.2 Stakeholders

Stakeholder	Role	Interest
Retention Team	Primary User	Identify and act on high-churn customer segments
Marketing Team	Secondary User	Design targeted retention campaigns
Business Analyst	Developer / Author	Build analysis and maintain the workbook
Senior Management	Decision Maker	Use KPIs to support budget and strategy decisions
Portfolio Evaluator	Reviewer	Assess BA and Excel analytical competencies

2.3 Assumptions

- The IBM Telco CSV dataset is the only data source — no external connections required
- The workbook user has Excel 2021or later installed on Windows or Mac
- Blank TotalCharges values (11 records) represent new customers and will be replaced with 0
- The project is for portfolio use only — not deployed in a production environment Data import, cleaning and transformation in Microsoft Excel

2.4 Constraints

- Analysis is limited to 7,043 rows — results may not generalize beyond this dataset
- All calculations must be Excel-native — no VBA macros, no external add-ins
- Workbook file size must remain under 20MB for easy sharing and GitHub upload

3. FUNCTIONAL REQUIREMENTS

3.1 Data Import and Cleaning

Req. ID	Requirement
FR-01	The system shall import the IBM Telco CSV file (7,043 rows, 21 columns) into a Raw_Data sheet without data loss
FR-02	The system shall replace 11 blank TotalCharges records with 0 in the Cleaned_Data sheet
FR-03	The system shall convert the TotalCharges column from text format to numeric format
FR-04	The system shall verify no duplicate customerID values exist; duplicates shall be flagged and removed
FR-05	The system shall confirm the Churn column contains only 'Yes' or 'No' values with no nulls

3.2 Feature Engineering

Req. ID	Requirement
FR-06	The system shall create a Churn_Flag column using: IF(Churn="Yes", 1, 0)
FR-07	The system shall create a Tenure_Bucket column grouping tenure into: 0–3, 4-6,7-12 months,1-2, 2+ Years
FR-08	Grouped monthly charges into price tiers : Low(<$30),Mid($30-$60),High($60-$90) and Premium($90 +)
FR-09	The system shall create a SeniorCitizen_Clean column using: IF(SeniorCitizen="Yes", 1, 0)

3.3 Pivot Table Analysis

Req. ID	Pivot Sheet	Row Fields	Values
FR-10	Churn By Contract Type	Contract	Churn_Flag → Count → Total CustomersChurn_Flag → Average → Churn Rate %
FR-11	Churn By Tenure	Tenure_Bucket	Churn_Flag → Count → Total CustomersChurn_Flag → Average → Churn Rate %
FR-12	Churn By Service	InternetService	Churn_Flag → Count → Total CustomersChurn_Flag → Average → Churn Rate %
FR-13	Churn By Payment	PaymentMethod	Churn_Flag → Count → Total CustomersChurn_Flag → Average → Churn Rate %
FR-14	Churn By Charges	Charge_Band	Churn_Flag → Count → Total CustomersChurn_Flag → Average → Churn Rate %

3.4 Charts and Visualizations

Req. ID	Chart Type	Data Source	Purpose
FR-15	Clustered Bar	FR-10	Share of Churn by Contract Type
FR-16	Line Chart	FR-11	Share of Churn by Tenure
FR-17	Clustered Bar	FR-12	Share of Churn by Service
FR-18	Clustered Bar	FR-13	Share of Churn by Payment
FR-19	Clustered Bar	FR-14	Share of Churn by Charges

3.5 Dashboard Requirements

Req. ID	Component	Type	Content	Purpose
FR-20	Dashboard Title	Text Header	"Customer Churn Analysis — IBM Telecom Industry"	Identifies the report subject
FR-21	KPI Box 1	KPI Card	Overall Churn Rate — 26.5%	Shows headline churn metric
FR-22	KPI Box 2	KPI Card	Highest Risk Segment — Month-to-month (42.7%)	Highlights worst contract type
FR-23	KPI Box 3	KPI Card	Biggest Churn Driver — Electronic Check (45.3%)	Highlights worst payment method
FR-24	KPI Box 4	KPI Card	Total Customers — 7,043	Shows total dataset size
FR-25	Chart 1	Clustered Bar	Churn Rate by Contract Type	Compares churn across contract types
FR-26	Chart 2	Line Chart	Churn Rate Across Customer Lifetime	Shows churn trend over tenure
FR-27	Chart 3	Clustered Bar	Churn Rate by Internet Service	Compares churn by service type
FR-28	Chart 4	Clustered Bar	Churn Rate by Payment Method	Compares churn by payment type
FR-29	Chart 5	Clustered Bar	Churn Rate by Monthly Charge Band	Compares churn by price tier

3.6 KPI Definitions

KPI	Excel Formula	Location in Dashboard	Expected Value
Overall Churn Rate	=COUNTIF(Churn,"Yes")/COUNTA(customerID)*100	KPI Box 1 — Top Lef	26.5%
Highest Risk Segment	Identified from Pivot Table — Churn by Contract	KPI Box 2 — Top Right	Month-to-month (42.7%)
Biggest Churn Driver	Identified from Pivot Table — Churn by Payment	KPI Box 3 — Bottom Left	Electronic Check (45.3%)
Total Customers	=COUNTA(customerID range)	KPI Box 4 — Bottom Right	7,043

4. NON-FUNCTIONAL REQUIREMENTS

Req. ID	Category	Requirement
NFR-01	Performance	All 5 Pivot Tables shall refresh in under 5 seconds when Raw Data is updated on standard hardware
NFR-02	Usability	The Dashboard sheet shall be interpretable by a non-technical business stakeholder within 2 minutes without any guidance
NFR-03	Accuracy	All KPIs (Churn Rate 26.5%, Total Customers 7,043, Churned 1,869) shall match SQL query results within ±0.1% tolerance
NFR-04	Compatibility	Workbook shall be fully compatible with Excel 2021 and later versions on both Windows and Mac operating systems
NFR-05	Portability	File size shall not exceed 20MB and the workbook shall open correctly after being downloaded from GitHub without broken links or missing data
NFR-06	Documentation	All 4 derived columns (Tenure_Bucket, Charge_Band, Churn_Flag, SeniorCitizen_Clean) shall have clear header labels and the Raw Data sheet shall document all transformation logic
NFR-07	Consistency	All 5 charts shall use a consistent colour scheme — Red for high churn, Orange for medium, Green for low churn across all visualizations
NFR-08	Completeness	Dashboard shall display all 5 required charts (FR-15 to FR-19) and 4 KPI boxes simultaneously without scrolling on a 1080p screen

5. Expected Key Findings (Portfolio Insights)

Based on the IBM Telco dataset, the analysis is expected to reveal the following business insights — demonstrating analytical depth for portfolio evaluation:

- Month-to-Month contract customers churn at a significantly higher rate (~42%) compared to 1-Year (~11%) and 2-Year (~3%) contract holders.
- Customers in the first 0–3 months of tenure have the highest churn probability, making early engagement the most critical retention window.
- Fiber Optic internet service customers churn more than DSL customers, despite (or because of) higher charges — suggesting service quality or price sensitivity concerns.
- Electronic check is the payment method most associated with churn, while credit card and bank transfer customers show lower churn rates.
- High-charge (Premium $90+) customers who are on Month-to-Month contracts represent the highest financial risk segment.
- Senior citizens represent a smaller segment but show a disproportionately higher churn rate, indicating a demographic-specific retention opportunity.

6. Document Sign-Off

Role	Name	Date
Author (BA)	Laxman Sharma	March 2026
Reviewer	— (Portfolio)	March 2026